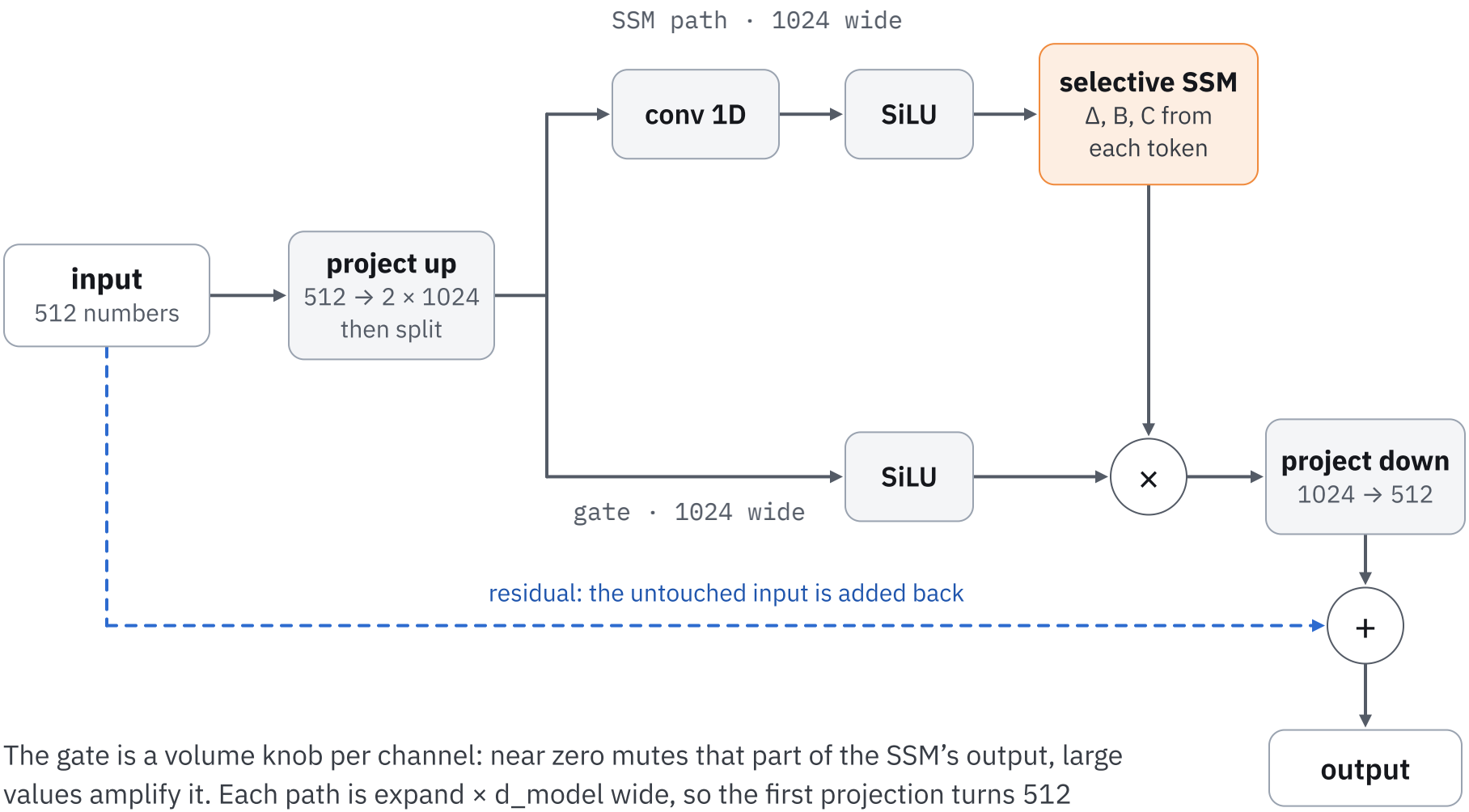
FIGURE 7

Inside a Mamba block (token width 512, expand × 2)



The gate is a volume knob per channel: near zero mutes that part of the SSM's output, large values amplify it. Each path is expand × d_model wide, so the first projection turns 512 numbers into 2 × 1024 = 2048.